

Depot

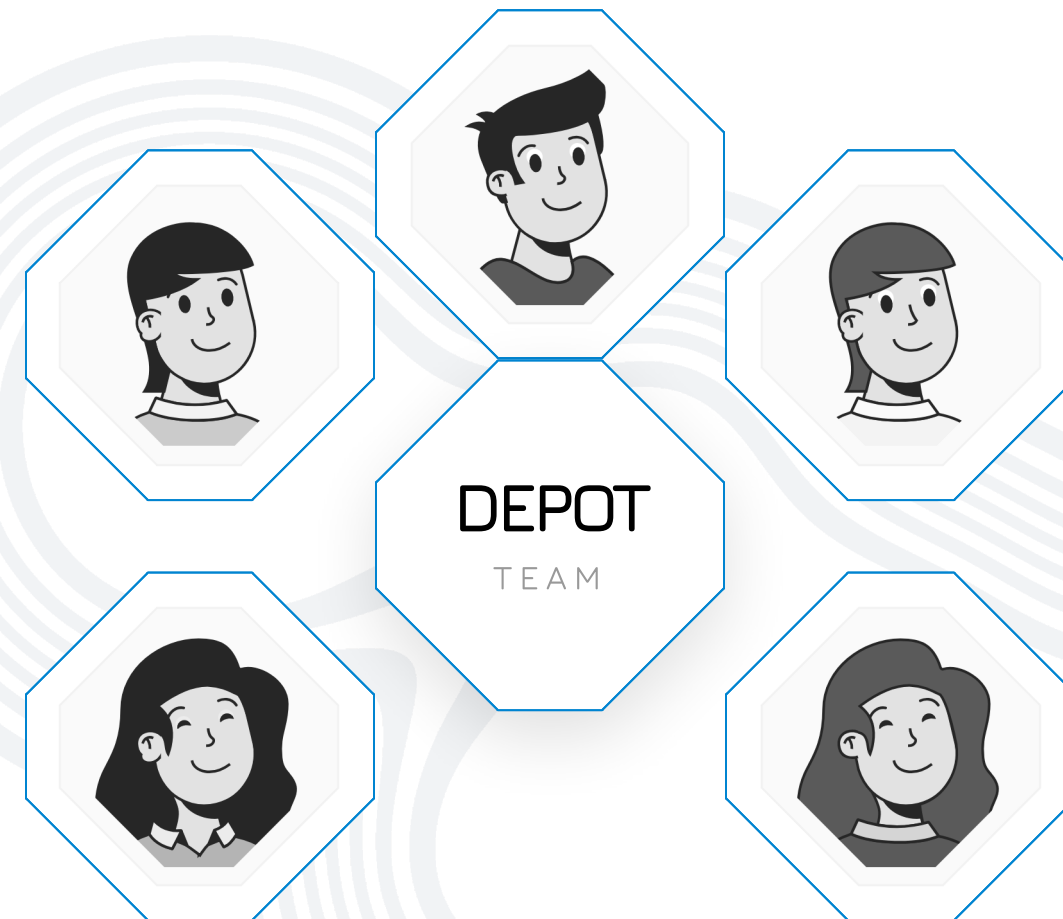
An AI-powered infrastructure architect,
from code to cloud.

TEAM

Ahsan Riaz
Muhammad Waleed
Muhammad Faizan Anwar

ADVISORS

Hira Anwar
Ayesha Hakim



Building apps is fast now. Shipping them is the bottleneck.

BUILDING IS FAST

~95%

of the code was written by AI in a quarter of Y Combinator's Winter 2025 start-ups.

Writing the product is no longer the hard part.

TechCrunch, March 2025 [1]

SHIPPING IS THE BLOCKER

19.36%

of infrastructure-code tasks were passed by GPT-4, against 86.6% of general coding tasks.

AI writes the app, but not the infrastructure that runs it.

laC-Eval benchmark, NeurIPS 2024 [2]

SHIPPING IS THE BLOCKER

150%

average yearly growth in infrastructure-as-code questions on Stack Overflow, 2011–2022.

Setting up infrastructure is a growing source of difficulty.

Study of Stack Overflow questions, ESEM 2023 [3]

Working code still leaves four decisions.

01

What it needs

02

Where it runs

03

What it costs

04

How to change
it safely

Depot answers all four, **and explains, records and approves every change.**

Each tool solves part of it. Someone still decides where it runs.

	Vercel managed platform	Flightcontrol bring-your-own-cloud	Depot proposed
Starts from the code	Plus platform config	Repo plus config file	Ordinary repository
Picks where it runs	The platform only	AWS only	Compares every option
Asks before it acts	No approval gate	Deploy only	Signed, risk-tiered

Built for builders without an infrastructure engineer.



AI-assisted builders

A working product, but little experience of running one.



Solo full-stack developers

Outgrowing a managed platform: stay, or move to a cloud?



Small teams and agencies

Two to ten people, with no DevOps staff.

Three objectives.

What a user can do with Depot.

01

Deploy any supported repository

Point Depot at a repo. It reads the app, picks the target, prices it, provisions it.

02

Check health, metrics and cost

History, health, logs, spend and rollback in one place.

03

Move a live service, not just its code

Cutover planned, priced and rehearsed before the switch.

Research sits inside all three: comparing language models on extraction, and measured guardrails for the agent loop.

What Depot actually does.



Application Model

Every fact marked detected, inferred or user-supplied.



No stored keys

Bounded roles, federation, or one scoped token per target.



Architect

Picks among reviewed patterns, ranked by cost and effort.



Delivery and operations

Deploy on push, rollback, logs, cost, alerts, savings.



Safety gate

Validate, saved plan, policy, cost and role check.



Agent and incident loop

Changes by conversation; alarms end in a fix a person approves.

One session, from working code to live.

STEP 1

Connect

GitHub, GitLab or an
archive

STEP 2

Answer

Up to eight
questions

STEP 3

Review

Target, architecture,
monthly cost

STEP 4

Approve

Signed and role-
checked

STEP 5

Run

Deploy on push, roll
back, watch cost

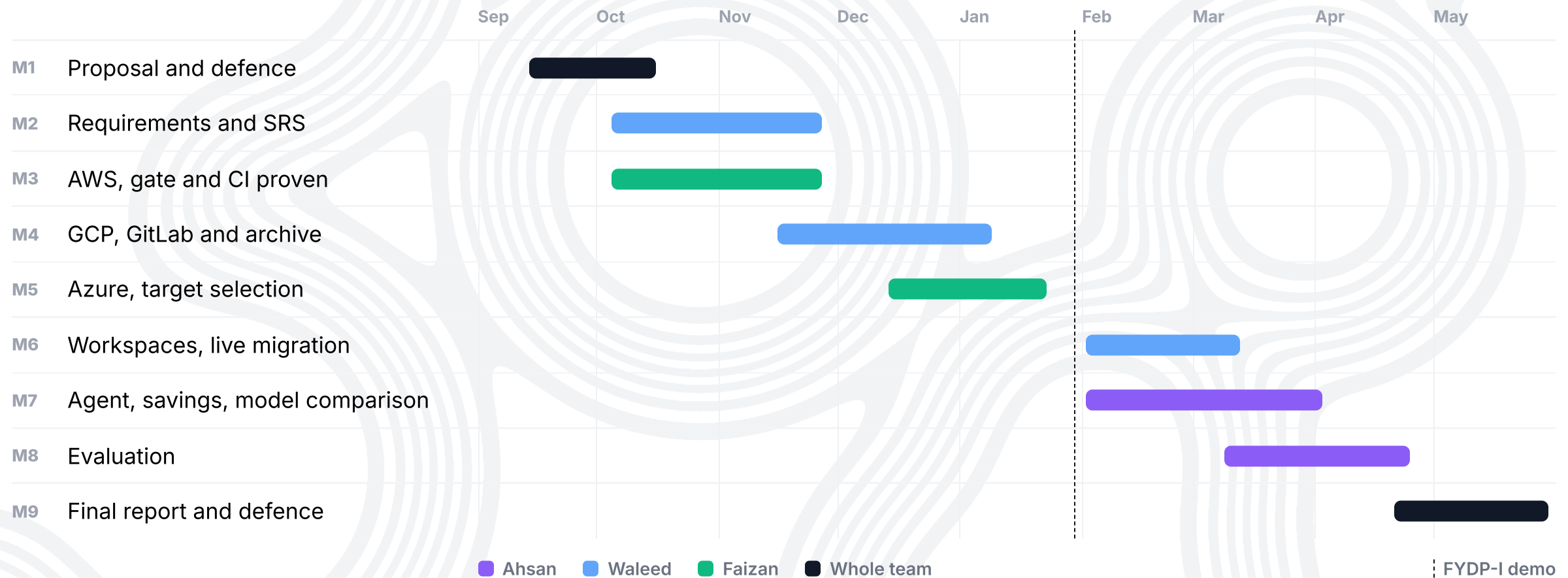
What Depot covers.

IN SCOPE

- ✓ GitHub, GitLab or an archive
- ✓ Apps, APIs, static sites, workers
- ✓ AWS, GCP, Azure
- ✓ Short-lived credentials
- ✓ Approval before every change

Extended scope only if M8 is met early: guided mode, Oracle and Yandex Cloud, voice, preview environments.

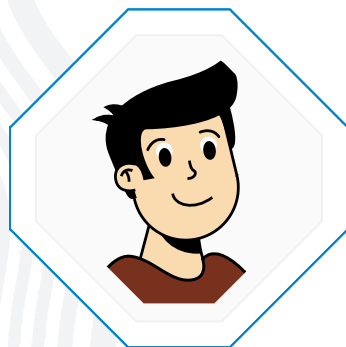
Three clouds first, then everything else.



Team planning targets; dates will follow the official defence schedule.

The Architects

Three owners. No member reviews only their own work.



Ahsan Riaz

Application Model, agent mode,
incident loop, research

01

02

03

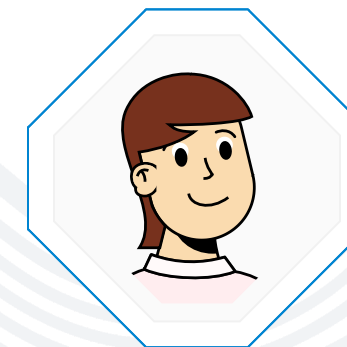


Muhammad Waleed

User-facing work, team workspaces,
managed platforms

01

02



Muhammad Faizan Anwar

Architect, cloud adapters, safety gate,
savings

01

03

Advisor **Hira Anwar** · Co-advisor **Ayesha Hakim**

Two goals we contribute to.

8 DECENT WORK AND
ECONOMIC GROWTH



SDG 8.3

Small-enterprise growth

Small teams run products on cloud accounts they own, at a known monthly cost, without hiring.

9 INDUSTRY, INNOVATION
AND INFRASTRUCTURE



SDG 9.5

Research and technological capability

An open benchmark, and views that explain why each architecture was chosen.

Sources.

- 1 I. Mehta, "A quarter of startups in YC's current cohort have codebases that are almost entirely AI-generated," TechCrunch, Mar. 2025.
- 2 P. T. J. Kon et al., "laC-Eval: A code generation benchmark for cloud infrastructure-as-code programs," NeurIPS Datasets and Benchmarks Track, 2024.
- 3 M. Begoug et al., "What do infrastructure-as-code practitioners discuss: An empirical study on Stack Overflow," ESEM, 2023.
- 4 Vendor documentation of the nine products compared, accessed Sep. 30, 2026.
- 5 OWASP Gen AI Security Project, "LLM06:2025 Excessive agency," 2025.
- 6 United Nations, "Goal 8" and "Goal 9," Sustainable Development Goals, accessed Sep. 30, 2026.

Full IEEE entries, with links and access dates, are in the proposal.

Thank you. Any questions?

• TEAM

Ahsan Riaz

Muhammad Waleed

Muhammad Faizan Anwar

ADVISORS

Hira Anwar

Ayesha Hakim

NUST SEECS · BS Computer Science